Append-Only Neural Memory: Storage, Addressing, and Growth
in a Depth-Recurrent Language Model

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September 28, 2026

Abstract

We study continual learning in BDH, a byte-level depth-recurrent language model [1] whose additive growth rule appends zero-initialized neurons while leaving all existing computation bit-identical. Our central finding separates three concerns that the continual-learning literature usually conflates. *Storage* is exact for the tested growth construction: under gradient-masked growth (each growth phase trains its new territory under a route-aware auxiliary loss, 0.9·own-slice-NLL + 0.1·full-width-NLL, Section 2.1), previously acquired computation remains bit-identical at the tensor level across every subsequent phase (verified independently on two hosts by two independent agent environments, across writing systems, including a cross-script transition). Throughout, *bit-exact* refers to equality of the preserved parameter tensors and buffers, not to identity of every floating-point output. *Serving* splits into two measured regimes: joint serving degrades because the summation readout accumulates contributions from all neurons (at the English-era checkpoint, 83% of the damage on the log scale, 62% in perplexity units, is reproduced by appending a single random, untrained block; across seven eras the same mechanism reproduces the majority of the damage in six, while the fraction itself is not a constant, range 0.60–1.47), while prefix-masked serving tracks acquisition within the measured routed-cost band (median +4.3% over the acquisition exit, worst case +8.0%), with zero drift across a full growth phase and 20/20 domains routing perfectly under an online-routing protocol, not a held-out split (per-domain Wilson floor $[0.91, 1.0]$ for 40/40). *Addressing* is the open problem: label-free likelihood selection achieves 20/20 correct prefix identification at every calibration budget from 4 KB to 1 MB, and a byte-level n-gram addresser reproduces the likelihood router on all 20 domains (8/8 held-out crops each; per-domain Wilson floor $[0.68, 1.0]$ for a perfect 8/8) given ∼4–8 KB of calibration text per domain—but cumulative-prefix NLL is not unimodal, so selection is robust yet not sublinear in territory count. Out-of-support detection requires two axes (relative advantage and absolute perplexity): byte-adjacent unseen languages have ratio ≈ 1 while cross-script inputs have catastrophic joint perplexity that inflates the ratio past the trained floor (the thresholds are fit and evaluated on the same n=6 probe set: a measured separation, not held-out validation). Cross-script probes show that routing correlates strongly with low-level byte statistics, and no linguistic reading is required to explain any observed decision: Chinese, Japanese, Hindi, and Inuktitut route 37–40/40 to the only two territories with substantial multi-byte training exposure (Cyrillic and Greek: 82% of characters are 2-byte-encoded, against 11.7% for the next-highest territory, a factor of 7), with zero linguistic kinship. A 20-phase fixed-capacity baseline shows that without growth, a 2k-step re-acquisition probe finds no measurable re-acquisition advantage over a fresh model (forgetting is not access loss). In the tested growth cell (Latin, Han, Devanagari), a Chinese territory grown on an English base, then Devanagari grown on Chinese, retains all three bit-exactly, routes perfectly, and serves each at acquisition quality.

1 Introduction

How should a trained network remain trainable? The continual-learning literature offers three families of answers: protect important parameters [7], store and replay experience [11], or allocate separate capacity per task [2, 3]. Each presupposes that preserving *parameters* preserves *computation*. In depth-recurrent architectures—which reuse one parameter block across all levels, obtaining depth without parameter count—the distinction is not academic. A frozen weight can still contribute to a changed computation, because the readout sums over all neurons, and growth adds neurons to that sum.

This paper takes BDH [1] as its instrument and reports a measurement arc that separates three concerns:

1. **Storage.** Under masked growth, is previously acquired computation physically preserved? We answer at the bit level for the tested configurations: yes, confirmations on two hosts by two independent agent environments, across writing systems, including a cross-script transition.

2. **Serving.** Given preserved storage, does the model still serve old tasks? We decompose the serving problem into *joint* (all neurons active) and *routed* (prefix-masked) regimes, and show the degradation in the former is 83% arithmetic on the log scale, 62% in perplexity units at the English-era checkpoint (reproduced by random untrained blocks; the mechanism holds in six of seven eras, the fraction does not) while routed serving remains within the measured cost band of acquisition and the hard mask preserves the old path by construction.

3. **Addressing.** Can the model find the right territory without an external language ID? We measure label-free selection, cheap input-side addressing, out-of-support rejection, and their scaling properties.

The resulting thesis is not that BDH solves continual learning. It is:

> *BDH provides an append-only neural substrate in which previously acquired computation remains physically intact while new computation is added. The central research problem is stable, efficient* ***addressing*** *of the accumulated territories—byte-separable in the regime measured here, and the open question for domains whose statistics are not—not destructive forgetting.*

Figure 1 summarizes the distinction used throughout this paper: storage is protected at the tensor path, serving exposes the shared summation readout, and addressing chooses the prefix that may contribute.

1.1 The measurement arc

Our experiments proceed from the simplest question to the hardest.

The forgetting baseline (Section 4). A fixed-capacity 100M BDH trained sequentially on 20 languages forgets catastrophically: by the final phase, nine of sixteen comparable languages serve at or below their English-only zero-shot level, two non-Latin scripts collapse by up to four orders of magnitude (bg $\times$12,086, el $\times$6,873), and a two-arm re-acquisition probe finds no measurable re-acquisition advantage over a fresh model at the tested budget (forgetting is not access loss).

The decay confound (Section 2.3). The original ladders carried a silent optimizer defect: AdamW's decoupled weight decay eroded gradient-masked weights by a closed-form per-phase factor,

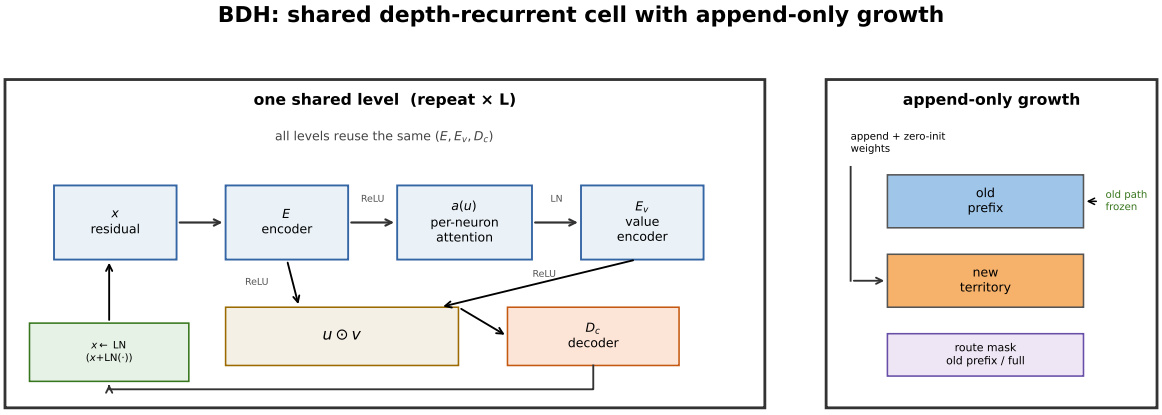


Figure 1: Schematic of the tested BDH cell and growth/addressing path. A single parameter triple (E, E_v, D_c) is reused across depth levels. Growth appends zero-initialized neurons and freezes the old path; route-aware training writes the new territory while an input-dependent mask selects which prefix may contribute at serving time. The diagram is architectural: the measured storage, routing, and readout claims are evaluated separately.

invisible to loss curves. We derive the closed form, verify it to four decimal places, and repair it with a step-end bit-exact restore.

Preservation under growth (Section 5). With the fix active, a 20-phase route-aware ladder (579M final) shows: position-dependent acquisition cost collapses, routing is perfectly diagonal (20/20 domains scored under all 20 prefix widths, 40 crops each; per-domain Wilson floor 0.91), retention remains within the measured acquisition-cost band (median +4.3%, worst case +8.0%, zero within-instrument drift across a full growth phase), and bit-exactness holds at every transition.

Readout mechanics (Section 6). Why does joint serving degrade if storage is bit-exact? A random-expansion control answers: appending one random untrained block costs English 4.1×; random expansion to full width reproduces 83% of the real damage at the English-era checkpoint (log scale; 62% in linear perplexity), and the mechanism reproduces in six of seven eras while the fraction does not (range 0.60–1.47); inert zero-blocks cost nothing. Every fixed readout operator in the seven families we test (gain rescaling, mass normalization, mixing, gradient-fitted per-territory gains) fails to repair it; we do not claim the space of input-independent operations is exhausted, and the measured pattern is about the tested families: the seven simple fixed readouts tested (two vacuous by construction) do not repair routed serving; a sufficiently expressive fixed map can itself implement input-dependent behaviour. The correct operation in our measurements is input-dependent selection.

Selection (Section 7). A label-free self-NLL selector identifies the correct prefix for all 20 domains at every calibration budget from 4 KB to 1 MB. A byte 1–4-gram logistic addresser reproduces the likelihood router perfectly under a class-balanced fit, given ∼4–8 KB of calibration text per domain. But binary search over prefix widths collapses: the cumulative-prefix NLL surface is not unimodal, so selection is robust but not sublinear in territory count.

Out-of-support detection (Section 7.3). A reject rule on two axes separates 20 trained languages from 6 unseen ones: routing advantage (joint/best-route ratio, $\geq 5\times$ for trained, $\approx 1\times$

for byte-adjacent unseen) and absolute best-route perplexity (within $\sim 10\times$ of the acquisition band). Cross-script unseen languages expose the ratio's degeneration: their joint perplexity is so catastrophic that the ratio inflates past the trained floor.

Cross-script generalization (Section 8). Routing correlates strongly with low-level byte statistics, and no linguistic reading is required to explain any observed decision: Chinese, Japanese, Hindi, and Inuktitut route 37–40/40 to the only two high-byte territories (Cyrillic, Greek). Chinese acquires from scratch at 2.69 ppl (best-val; test 2.48); a Devanagari phase grown on Chinese retains both bit-exactly and serves both at acquisition quality (full artifacts: `docs/reports/2026-09-11_cross-script-rejection-suite.md`). This is the tested cross-script cell (two growth transitions): it shows the construction remained functional across script families, not that every future transition is guaranteed.

Prior art (Section 9). We position each result against the literature: additive growth (Progressive Networks, PackNet, Piggyback, SupSup), task addressing (Expert Gate), consolidation (EWC/SI/MAS), replay (GEM), MoE routing (Switch/GShard), selective prediction (Chow, conformal), and production-scale conditional memory (DeepSeek Engram).

1.2 Pre-registered prediction and outcome register

The register below (Table 1) separates predictions fixed before their outcomes from later controls and from the measured limits of each claim. A failed or partial prediction is reported as such; it is not silently replaced by a post-hoc success.

Table 1: Pre-registered predictions, outcomes, and evidence scope.

ID	Pre-registered prediction	Outcome	Evidence / scope
P-FCS-1	Fixed-capacity sequential loading will produce catastrophic forgetting.	**PASS**; family-structured erasure is the measured form.	Fixed-capacity matrix; 20-domain protocol.
P-FCS-2	Acquisition remains comparable across phases, without a late-phase cliff.	**PASS**.	Acquisition-floor matrix; each phase remains within the reported band.
P-FCS-3	Backward interference will be dominated by the most recent phase.	**FAIL**; family structure is stronger and more specific.	FCS matrix; prediction falsified by family-dependent oscillation.
P5	Old masked segments and optimizer state remain protected at every tested growth transition.	**PASS**; standing integrity gate.	In-chain P5 checks; separate confirmations reported in the growth section.
H-decay-1	Step-end restore removes old-segment and routed drift.	**PASS**.	RA2b readout; 20/20 routing and zero p19–p20 routed drift.
H-decay-2	Joint serving recovers after the decay repair without a splice.	**PASS**; residual interference remains.	RA2b joint readout; the recovery claim excludes a no-op.
H-decay-3	Within-family acquisition gaps shrink markedly under the fix.	**PASS**.	RA2b same-host late-phase comparisons; position cost collapses.

ID	Pre-registered prediction	Outcome	Evidence / scope
P-R1	A tested fixed readout operator can repair growth damage.	**NEGATIVE**; no pass condition met.	P-R1 operator sweep; culling and fixed shifts fail or are vacuous.
P-R1b	A language-agnostic scalar reweighting can repair growth damage.	**REFUTED**.	P-R1b; calibration helps one language while damaging others.
P-R2	Width growth can reproduce most joint-serving damage without new learning.	**SUPPORTED**; 83% log-scale, 62% linear.	Random/inert/real expansion control; English-era result, mechanism across six of seven eras.
P-R3	A cheap byte-geometry addresser can reproduce the likelihood router.	**SUPPORTED after correction**; balanced fit reaches 1.00 in-support.	Initial 76% fit withdrawn after a composition-confound audit; balanced refit is the reported result.
P-R4	Conformal thresholds can be frozen for OOD rejection.	**NOT STARTED**.	Thresholds remain empirical and fit/evaluated on the same six-language probe set.
P-X1	Cross-script inputs have no trained-quality territory.	**PASS**.	Cross-script suite; all tested inputs remain far above the trained band.
P-X2	Cross-script routing concentrates on high-byte territories.	**PASS**; clean rerun strengthens it.	zh/ja/hi/iu-clean route 37–40/40 to bg/el; no Latin attraction after correction.
P-X3	A ratio-only OOD rule remains below the trained floor at maximum byte distance.	**PARTIAL**; the one-axis rule fails, while the two-axis rule separates.	Cross-script suite; the corrected two-axis rule uses ratio and absolute PPL.

2 Setup

2.1 Architecture primer

Table 2 fixes notation. One BDH level computes, with encoders $E, E_v \in \mathbb{R}^{n_h \times d \times N}$ and decoder $D_c \in \mathbb{R}^{(n_h N) \times d}$,

$$u = \mathrm{relu}(x@E), \qquad v = \mathrm{relu}\big(\mathrm{LN}(a(u))@E_v\big), \qquad x \leftarrow \mathrm{LN}\Big(x + \mathrm{LN}\big((u \odot v)@D_c\big)\Big),$$

where $a(\cdot)$ is per-neuron attention over positions of x: each latent neuron attends independently, there are no query/key/value projection matrices, and every operator above is coordinate-separable across neurons. Concretely, $a(u)$ scores positions with RoPE-rotated u via unnormalized causal linear attention (a masked $\mathrm{QR}\,\mathrm{QR}^\top$ product multiplying the residual x as values—no softmax, no value projection), and returns a residual-width vector per head. The only cross-neuron coupling is LayerNorm's [18] global statistics. One triple (E, E_v, D_c) serves all L levels. The attention block carries no trainable parameters (its only state is the RoPE [19] frequency buffer), which matters throughout: all learning lives in E, E_v, D_c, the token embedding, and the output head.

Table 2: Notation used throughout.

symbol	meaning
$d,\ n_h,\ N,\ L$	residual width, heads, neurons/head, depth levels
$x \in \mathbb{R}^d$	persistent residual state (shared across levels)
E, E_v, D_c	shared neuron operators (encoder, value encoder, decoder)
phase A, θ_A	a training stage and its parameters
$F_A,\ F'$	specialist map; extended (grown) map
$\mathcal{X}_A,\ \mathcal{T}_A$	old-task input set; reachable old-trajectory states
P_A	projection/mask selecting the old-phase contribution channels
territory b	the neurons appended by phase b (a width interval)

Growth appends zero-initialized neurons to the triple; existing neurons, the token embedding, and the output head are frozen. RoPE frequencies carry the neuron count in their exponent, so naive growth would silently rewrite every surviving neuron's phases; the growth path preserves the frequency prefix verbatim. (An implementation quirk, recorded for fidelity: the training script writes the new-frequency formula with Python's right-associative exponentiation, evaluating $2^{(16^r)}$ rather than $(2^{16})^r$; this affects only the frequencies of newly grown neurons—old neurons' frequencies are copied verbatim—and no preservation claim.) During a growth phase, a gradient mask zeroes the gradients of all pre-existing neurons, so new-phase updates write only the new capacity; a step-end restore additionally guarantees bit-exactness of the frozen path against optimizer side effects (Section 2.3).

Route-aware training couples the masked growth with a per-phase auxiliary loss: the phase minimizes NLL on its own data both under the *own-slice* mask (only the neurons grown in this phase, the width interval $[n_{\text{old}}, n_{\text{new}})$, are active) and over the full width, with weight α on the own-slice term ($\alpha = 0.9$ throughout; the training logs print this as "prefix"). "Prefix" elsewhere in this paper always means a cumulative width $[0, k)$ at serving time—a different object. The effect, measured below: the phase writes its knowledge into its own territory while the shared readout stays calibrated enough that prefix selection recovers it.

2.2 Protocol

All experiments use byte-level BDH on Europarl [24] and the cross-script corpora below: the vocabulary is the 256 byte values, sequences are raw byte streams, and a phase is a ∼30 MB byte stream. Growth phases initialize from the previous endpoint (weights only; fresh optimizer). Evaluation is per-domain held-out perplexity under a fixed cold random-crop protocol (block 512, 40 crops, generator 1234), always within-corpus. Where we compare joint and routed serving, the routed measurement uses the likelihood router of Section 7 and the joint measurement uses the full width on the same crops. This is a controlled multilingual accumulation trial over one parallel-corpus ecosystem: it is closer to continual pretraining than to independent-task continual learning, and is read as such. All results are reported for the dense-ReLU configuration (k_sparse_ratio $= 0$); ratio-based top-k sparsification under growth is not width-invariant and is outside these claims.

Hardware: one RTX 4090 (24 GB) and one NVIDIA GB10 (gx10, 121 GB); scale is ∼100M parameters at the base width (×128) and ∼579M at the final ladder width (×736). Every number in this paper traces to a committed artifact in the project repository (`https://github.com/Saga-AI-Labs/bdh-cl`; reports, plans, data files, and pre-registration headers in the training scripts); the project's internal message bus is the working coordination layer, and where a claim cites its provenance, this paper names the repository artifact.

setting	value
model	byte-level BDH, $d = 512$, $n_h = 8$ (64 neurons per head per unit multiplier)
training sequence	512 bytes, batch 4 at width up to ×192 (20.48M tokens per phase), batch 1 beyond (5.12M tokens per phase)
optimizer	AdamW, lr $= 10^{-3}$, weight decay 0.1, $\beta = (0.9, 0.95)$, grad-clip 1.0
schedule	warmup 1000, cosine to min lr $= 10^{-4}$ over 10,000 iters
phase budget	10,000 iters on a ∼30 MB byte stream per language
objective	plain next-token NLL (fixed-capacity floor); 0.9·own-slice-NLL +0.1·full-NLL (growth phases 2–20)
growth	+32 neurons/head per phase from a ×128 base (+2048); $\alpha = 0.9$; attention parameter-free (RoPE buffer prefix-preserved)
training data	Europarl v7 byte streams; byte vocabulary 256
evaluation	cold random crops, block 512, 40 crops, generator 1234
matrix eval	`lang_eval.py` with eval batch pinned to 1 (all 400 cells)

2.3 The decay confound, closed

The original growth ladders carried a silent optimizer defect that we discovered and repaired mid-project; every pre-repair number in this paper carries its consequence, and we state the mechanism once, here.

Mechanism (derived). AdamW's [17] decoupled weight decay multiplies every parameter that has a gradient by $(1 - \mathrm{lr}_t \cdot \mathrm{wd})$ each step. A gradient mask that zeroes the frozen weights' gradients does *not* stop this: a zero gradient still counts as a gradient, so masked weights shrink by the schedule product

$$c \;=\; \prod_{t=1}^{T} (1 - \mathrm{lr}_t \cdot \mathrm{wd}), \qquad p_{\mathrm{exit}} = c \cdot p_{\mathrm{entry}}$$

independent of data, loss, or routing as a property of the schedule; this clause is derived rather than separately tested, since no experiment varies the data at a fixed schedule. Under the default lr schedule (warmup 30, decay 300, plateau at 10^{-4} for 97% of steps) the per-phase factor is $c = 0.8927$; under the cosine schedule (warmup 1000, decay 10000) it is $c = 0.5798$. The f32 realization of the same product sits 1.36×10^{-8} below the exact-precision value per plateau step (float rounding of $1 - \mathrm{lr} \cdot \mathrm{wd}$), which over the ∼9,700 plateau steps accumulates to about -1.3×10^{-4} and reconciles the measured 0.892636 with the exact-precision 0.892752.

Verification. The closed form matches measured per-segment scale factors on every checkpoint transition we tested: residuals $\sim 10^{-5}$ across 1-, 4-, 18-, and 19-phase intervals (Figure 2 plots the two regimes against the measured factors), three independent instruments (torch c-fits, optimizer-moment census, weight-atlas spectral fingerprints) agree to four decimals. In the other direction, the leak is invisible where it matters most: single-phase acquisition is unaffected (bounded regime nulls at $\leq 2.5\%$, within the 2–4% seed floor), because the erosion acts over *many* phases.

Repair. A step-end restore re-establishes bit-exactness of the masked path after every optimizer step. The confirmation ran on two hosts, by two independent agent environments, with two independent verification codebases: old-segment bit-identity across growth phases, $c = 1.000000$ exactly. All fixed-regime numbers below (the RA2b chain, cross-script stages) run under this repair. The cost is measured: 8.7 ms per GiB restored at the final width (2.06 GiB snapshot, 17.9 ms against 3038 ms/step, 0.59% of step time; the snapshot is held resident for the phase, +2.06 GiB, which is the binding constraint on small-memory hardware).

Precedent. The same failure signature—routed-expert norms falling toward zero under AdamW+weight decay while evaluations stay normal—has been observed independently in production MoE training (Marin project tracker, thread 8818) [22], suggesting a failure class, not an idiosyncrasy of our setup.

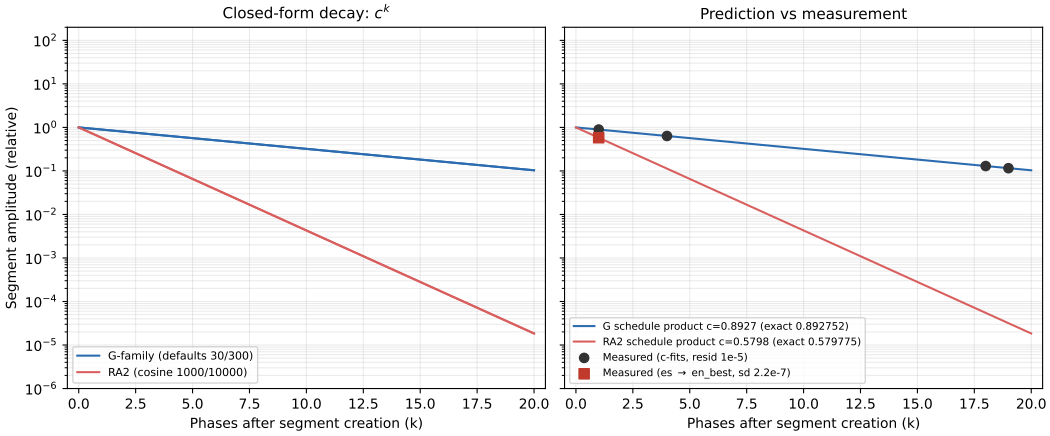


Figure 2: Optimizer decay regimes used to audit the confound. The closed-form schedule product predicts the measured per-segment scale factors; the RA2 and default schedules differ in the resulting decay rate.

3 Theory: storage is exact; serving and addressing are not

The theory of rev 3 carries over unchanged in its core; we restate the two load-bearing results and refer the proofs to Appendix B. The new empirical sections then measure what the theory predicts: growth constructs exact storage, the readout breaks serving, and selection repairs it.

One clarification, because it is easy to misread the dissociation result: the construction of Theorem 1 is realized *after* training, by a suffix whose parameters are non-zero. The zero-initialized suffix of the growth convention measures the *untrained* case—which is exactly why the A−C gap in the random-expansion control quantifies that convention's protective value, rather than contradicting the theorem.

Theorem 1 (dissociation)**.** PROVED. *Consider the class of depth-recurrent maps whose level update is invariant under constant shifts of the state, $T(h + c\mathbf{1}) = T(h)$ ($\mathrm{LN}(h + c\mathbf{1}) = \mathrm{LN}(h)$-style normalization included). In this class there exist extended maps F' of the form $F'(h) = F_A(h) + c\,\mathbf{1}$ with a* per-level *shift: the constant $c \neq 0$ is added at each level invocation of the depth recursion. The shift does not accumulate across depth—the correct formula is $F'^{\ell}(x) = F_A^{\ell}(x) + c\,\mathbf{1}$, not $+Lc\,\mathbf{1}$—but it persists at every depth $\ell \geq 1$: the level map cannot see the offset, so the divergence $c\,\mathbf{1}$ is re-established after every iteration. Any such F' whose parameter set contains θ_A unchanged still has old-task trajectories that differ from the specialist's at every depth. Parameter isolation imposes no bound on computation isolation.*

Implementation scope. The theorem is stated for the abstract shift-invariant class, not as a claim about the literal level code. In the implementation a level *ends* with LayerNorm, $x \leftarrow \mathrm{LN}(x + \mathrm{LN}((u \odot v)@D_c))$, and the encoder read inside the level is not re-normalized, so the literal level map does not satisfy the shift-invariance hypothesis, and its zero-mean output cannot realize $F'(h) = F_A(h) + c\mathbf{1}$ with $c \neq 0$ by appended neurons alone. The theorem's role is the structural one: parameter isolation alone rules out neither altered old-task computation nor its opposite. In the literal architecture, appended computation perturbs old-task serving through the readout sum inside $\mathrm{LN}(x + \mathrm{LN}(\cdot))$—the mechanism measured in Section 6 (Remark 2).

Theorem 2 (exact-isolation criterion)**.** PROVED. *Let $\mathcal{T}_A = \bigcup_{0 \leq \ell < L} \{F_A^{\ell}(x) : x \in \mathcal{X}_A\}$ be the reachable old-trajectory set of the preserved specialist. Then $F'^{\ell}(x) = F_A^{\ell}(x)$ for all $x \in \mathcal{X}_A$ and*

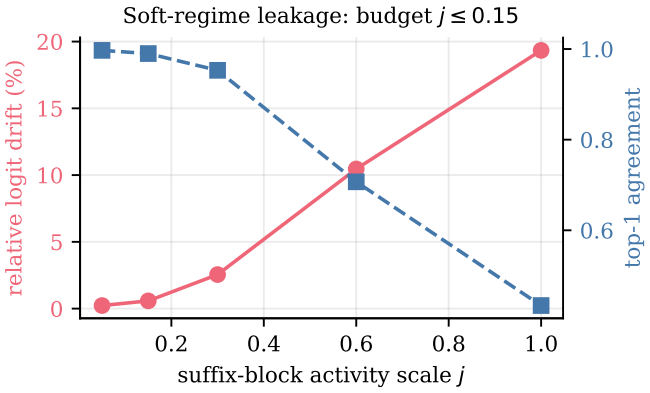


Figure 3: Measured soft-regime leakage from a suffix block into old-language serving: 16 English evaluation windows, with the appended ES suffix block's contribution scaled by an activity multiplier $j \in [0, 1]$ (a magnitude knob on the appended block, not a learning rate). Relative logit drift and top-1 agreement vary with j; the operating budget quoted in the plot title, $j \leq 0.15$, is where top-1 agreement stays $\geq 99\%$. The hard endpoint (j=0) remains the only exactly measured preservation mechanism. This figure is evidence for the scope of the theory, not a replacement for the proof.

all $0 \leq \ell \leq L$ ***iff*** $F'(z) = F_A(z)$ *for all* $z \in \mathcal{T}_A$. *A sufficient structural decomposition: with* $H_A = \operatorname{im} P_A$ *containing* $\mathcal{T}_A$ *and* $F_A(H_A) \subseteq H_A$,

(C1) invariance*:* $F'(H_A) \subseteq H_A$*;*

(C2) restriction equivalence*:* $P_A F' P_A = P_A F_A P_A$ *on* H_A.

Neither condition alone suffices.

Corollary 1 (prefix growth constructs the structure)**.** PROVED. *Under hard suffix masking, the grown map restricted to masked forwards satisfies (C1)–(C2) identically. Hard selection therefore reproduces specialists exactly.*

Remark 1 (soft activity, measured not proved). MEASURED/HEURISTIC. Under *soft* activity, LayerNorm's global statistics break (C1) at first order in the suffix magnitude (Figure 3). This is the reading our readout measurements support rather than a proved corollary: the argument is not written out here, and Section 6, which uses it, treats the consequence as measured.

Remark 2 (readout arithmetic, measured). MEASURED (Section 6). With $k_\text{sparse}_\text{ratio} = 0$, the readout sums over all N ReLU candidates, and growth adds terms to an existing sum with bit-frozen old weights. Theorem 1 predicted this class of failure abstractly; the random-expansion control measures it at the English-era checkpoint: one random block costs 4.1×, and 83% (log scale; 62% in linear perplexity) of the real damage needs no learning at all.

Lemma 1 (zero-forcing)**.** PROVED. *If suffix blocks are individually gated,* $f_b \mapsto g_b(x) \odot f_b$*, then exact preservation on* $\mathcal{X}_A$ *forces* $g_b(x) = 0$ *wherever* f_b *is nonzero along the preserved trajectory (blocks preserved sequentially, phase by phase, as in the growth protocol; concurrent summed suffix activity is pinned only up to cancellation). (The specialist term is un-gated; only suffix activity is constrained.) Gate freedom lives on never-active coordinates and new-task inputs.*

Proposition 1 (a cross-coupled ReLU counterexample to exact soft gating)**.** PROVED *(counterexample class). For non-affine updates with cross-coupling, the preservation constraints form a functional equation over the trajectory with no solution in an open gate range. Concretely: in the two-dimensional ReLU system below with gates* $g_1, g_2 \in [0,1]$*, cross-couplings* $\varepsilon, \delta > 0$*, self-gain* $\gamma \geq 0$*, and identity specialist* $F_A = \mathrm{id}$*, preserving the trajectory of every* $x_0 = (a, 0)$*,* $a > 0$*, for two steps reduces to* $(1+g_1)^2 + g_1 g_2 \varepsilon\delta = 1$ *together with* $g_2(2+g_1+\gamma g_2) = 0$*, whose only solution is* $g_1 = g_2 = 0$*. Exactness collapses to the hard endpoint; soft regimes can only bound.*

Proposition 2 (depth amplification)**.** PROVED *(standard recursion). With local injections* δ_ℓ *and level-map Lipschitz constant* $L_{F'}$ *(of the composed level update; Remark 3 measures exactly these whole-map gains),* $\|e_L\| \leq \sum_j L_{F'}^{L-1-j}\delta_j$*. Uniform-in-depth bounding additionally requires contraction (*$\rho < 1$*) on the interference dynamics, realized as finite differences in Appendix B.7 and measured in Remark 3; the uniform bound is vacuous at the tested depths, and the measured directional gains do not support the premise.*

Remark 3 (expansiveness: measured, not proved-impossible). MEASURED. Directional spectral-norm proxies of trained BDH levels at old-input states are 1.05–1.89 (median per level). This is evidence *against* the contractivity premise of Proposition 2 for the tested models—uniform contraction-based bounds are unsupported—and *not* an impossibility theorem: isolated stable directions may coexist with expansive medians.

Corollary 2 (selector vs. creator, scoped)**.** *Within the frozen-path projection mechanism studied here, gates do not create invariant computational structure; they select structure already present in the frozen operator. In broader architectures a learned gate can alter effective computation; that possibility is outside this mechanism class.*

4 The forgetting baseline: what happens without growth

Before any claim about growth, we measure what a fixed-capacity BDH does under a pure sequential load: 20 languages, no growth, no masks, no replay, one set of weights overwritten phase by phase, and no route-aware auxiliary loss (the fixed-capacity floor uses the plain next-token objective, while the growth ladder couples training to a prefix-masked loss; Section 2.2). This is the floor every mechanism must beat, and it was missing from the literature side of our own project until we asked for it explicitly (pre-registered as P-FCS-1–3).

4.1 Design

Fixed capacity ×128 (∼100M), same sequence and evaluation protocol as the growth ladder (en, es, pl, fr, de, cs, da, pt, fi, hu, bg, it, et, el, sk, sv, ro, nl, sl, lt), 10k steps per phase at batch 4 (20.48M tokens per phase in every one of its 20 phases, confirmed against `scripts/ladder_fixedcap.sh`), fresh optimizer per phase (weights restored via `--init-from`). After every phase, all 20 domains are cold-evaluated (block 512, 40 crops, generator 1234), yielding a 20×20 perplexity matrix whose rows are training prefixes. The full matrix is `2026-09-10_fcs_matrix.csv` under the project reports data directory. Note the token-budget asymmetry with the growth ladder: FCS is batch 4 throughout, while the ladder switches to batch 1 at phases 5–20 (Section 2), so FCS-vs-ladder is not token-matched—the ladder used 4× less data per phase in 16 of its 20 phases. The asymmetry favours the ladder (comparable acquisition on a fraction of the data) but any cross-phase trend in the ladder must not be read as coming from a single data budget.

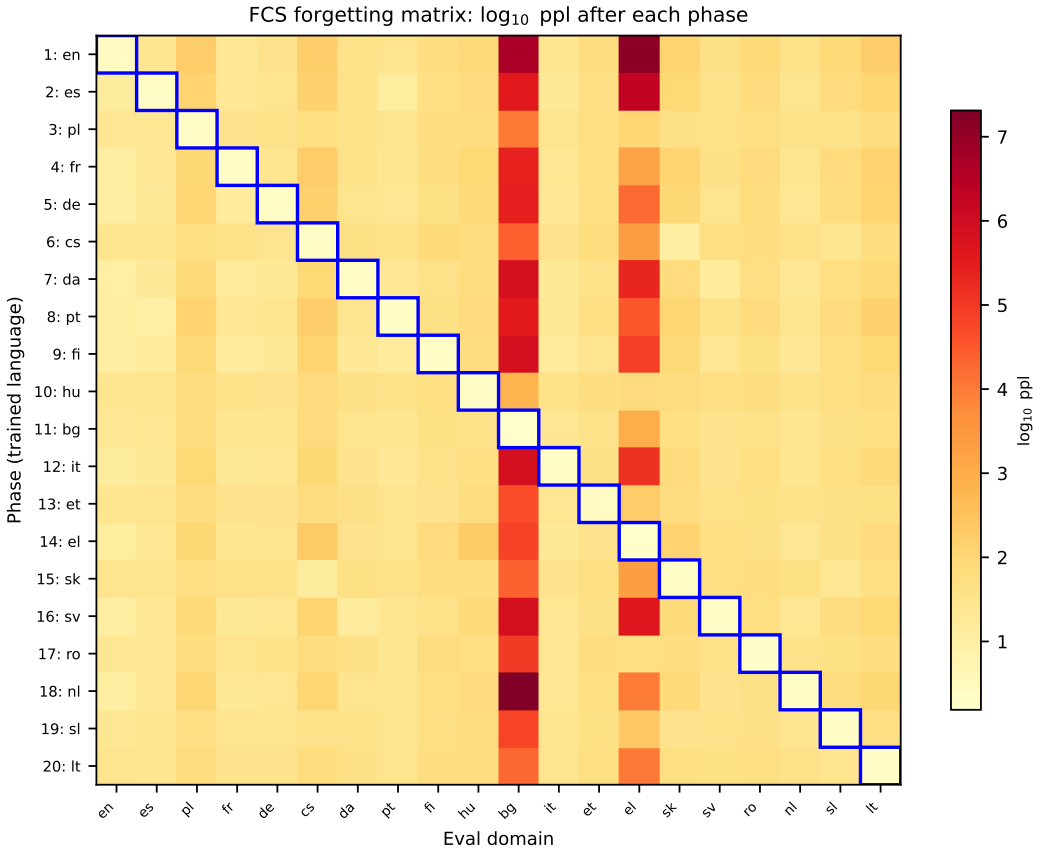


Figure 4: FCS forgetting matrix ($\log_{10}$ ppl), 20×20. Each row is the 20-domain cold eval after that phase; blue box marks the diagonal. Latin-script languages fall to their English-only zero-shot level (nine of the sixteen zero-shot-comparable domains fully displaced, seven partial retention; en, lt, bg, el excluded); bg/el collapse by up to four orders of magnitude at row 20 (bg $\times$12,086, el $\times$6,873 relative to acquisition); family-structured oscillation survives throughout.

4.2 Results

Acquisition is never the constraint (P-FCS-2 PASS). Every language acquires at its own phase between 1.54 and 2.29 ppl—including the 20th (lt, 2.13). Fixed 100M capacity saturates for no single language. Notable inversion: the non-Latin scripts acquire *best* under full overwrite (bg 1.54, el 1.59) while they were the worst acquirers under growth+selection (5.86, 5.99 in the fixed-regime ladder)—under full overwrite the entire model serves the current language, so there is no protected capacity to fight over.

Forgetting is family-structured, not total (P-FCS-1 PASS, corrected). Row 20 (after all 20 phases) splits the sixteen zero-shot-comparable domains by family: *nine fully displaced*—serving at or above their English-only zero-shot level (es, fr, de, it, pt, da, sv, nl, fi—the Romance/Germanic cluster plus Uralic *fi*, whose byte profile sits with that group; strongest zero-shot transfer, displaced back to exactly what English alone transferred); *seven partial retention* (pl 0.34×, sl 0.35×, cs 0.37×, sk 0.43×, ro 0.61×, hu 0.78×, et 0.87× their zero-shot); and the two non-Latin scripts collapsed up to four orders of magnitude at row 20 (bg ×12,086 → 18,613, el ×6,873 → 10,928; the twentieth domain, *lt*, is the language trained in row 20 itself and serves there at its acquisition value, 2.13, retained rather than displaced). The family axis that governs interference also governs survival.

Interference is not recency-structured but family-structured (P-FCS-3 FAIL—replaced by a stronger finding). Pre-registered prediction: the most recent phase dominates backward interference. Measured (Figure 4, en column): en serves at 10.5–13.6 ppl after the Romance/Germanic phases (es, fr, de, da, pt, it, sv, nl) and after *fi* (11.31), but rises to 22.1–29.1 after the Slavic phases (pl, cs, sk, sl) and their band-mates *hu*, *et* (26.20), and Romance *ro* (22.48), with a Germanic phase *partially restoring* en after a Slavic one (29.1 → 11.3). The misfits are the finding, not a labeling slip: *fi* behaves like the Romance/Germanic group while its Finnic relative *et* sits in the high band, and *ro* behaves like the Slavic band—byte statistics, not genealogy, predict the interference bands. The mechanism: shared Latin-script byte statistics act as implicit replay. This is the family-geometry finding at its cleanest.

Forgetting is destruction, not access loss (two-arm probe). Fine-tune on bg for 2k steps from two bases: the row-20 endpoint (bg trained 19 phases ago, serving 18,613) reaches 1.67; a fresh English-only base (bg never trained, zero-shot 4.9M) reaches 1.70 under the identical budget. $\Delta = 1.8\%$, below the 2–4% seed floor: the once-trained, 19-times-overwritten state contributes *nothing measurable* to re-learning speed. Resolution caveat (accepted in review): fixed-2k endpoints cannot separate "destroyed" from "intact but slowly re-accessible"; the claim is resolution-bounded. Scope: fixed capacity only—the growth regime is the opposite (bit-exact preservation, Section 5).

A first-class baseline, not only literature. We did not run this floor against every classical competitor at matched compute, and we do not claim to have. But our own replay-in-training result belongs in the evidence, not only in the prior-art discussion: a replay arm (H1p) reached joint parity with the growth construction at +27% additional training budget in the fixed-capacity regime, establishing the concrete price at which replay buys back what overwrite destroys. Replay pays data and compute; growth pays serving and addressing. The comparison is stated and sourced in Section 9.4, not re-run here.

5 Preservation under growth: the fixed-regime ladder

The complementary experiment: the same 20-language sequence, but each phase *grows* the model by +32 neurons/head and *masks* all pre-existing gradients, under the decay repair of Section 2.3. Pre-registered predictions H-decay-1/2/3 were written before any number existed.

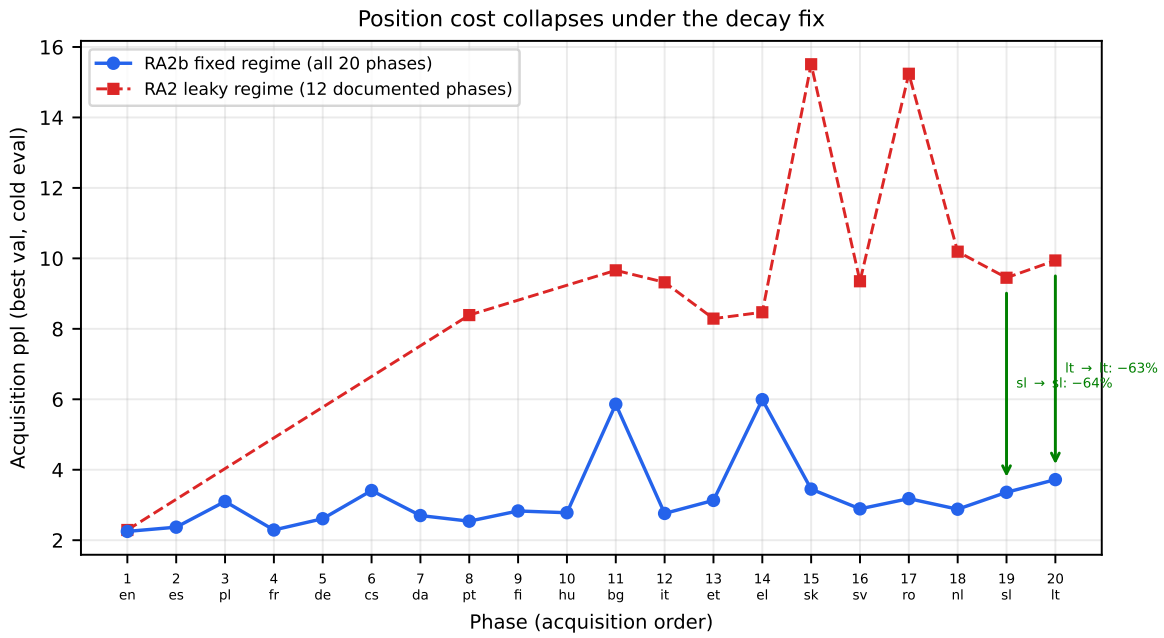


Figure 5: Ladder acquisition. RA2b (fixed regime, blue, all 20 phases): 2.25–5.99 band, position cost gone. RA2 (leaky regime, red, 12 documented phases): acquisition tracked ladder position. The x-axis is the phase index: RA2's mid-ladder language order (phases 6–18) differed from RA2b's, so only phases 1–5, 19, and 20 share the bottom labels, and the red squares are RA2's own per-phase values under its own ordering. Green arrows: same-position same-host cells lt 9.94 → 3.72 (−63%), sl 9.45 → 3.36 (−64%). Residual spread is alphabet difficulty (bg/el), not chain state.

5.1 Acquisition: position-spread collapses

Final-chain best-val perplexities span 2.25–5.99 (Figure 5), and the spread that characterized the leaky era is gone: same-position same-host cells lt (phase 20) improved from 9.94 to 3.72 and sl (phase 19) from 9.45 to 3.36. Late-position languages now acquire in the same band as early ones (sv 2.89 at position 16, nl 2.88 at 18)—*better* than the leaky ladder's early-position languages. What remains above the Latin band is alphabet difficulty, not chain state: bg 5.86, el 5.99, lt 3.72.

H-decay-3 PASS: the within-family acquisition gaps of the leaky era were driven substantially by decay erosion of earlier segments' serving capability at acquisition time.

5.2 Routing: perfectly diagonal

The 20-way routing diagnosis on the final chain scores every domain under all twenty prefix widths (40 crops each, 800 crop-width cells) and resolves **20/20 domains to their own prefix**—every domain, every crop, its own training width (the per-domain Wilson floor for a perfect 40/40 is $[0.91, 1.0]$; crops within a domain are not independent, so the raw 800 should not be read as 800 independent trials). Under the leaky chain the same instrument showed 36/40 for fi (four crops lost to its Estonian neighbor) and family wanderings (cs/pl→sk, Romance→ro); none of that remains. The family structure that governs *error* modes (FCS oscillation, cross-script attraction) disappears when every territory exists: routing is exact. Because the router selects on the first 128 tokens of a crop and the serving evaluation uses the remaining 384 tokens of the same crop, this is an online-routing protocol, not a held-out split. At 20/20 domains with 40/40 crops each,

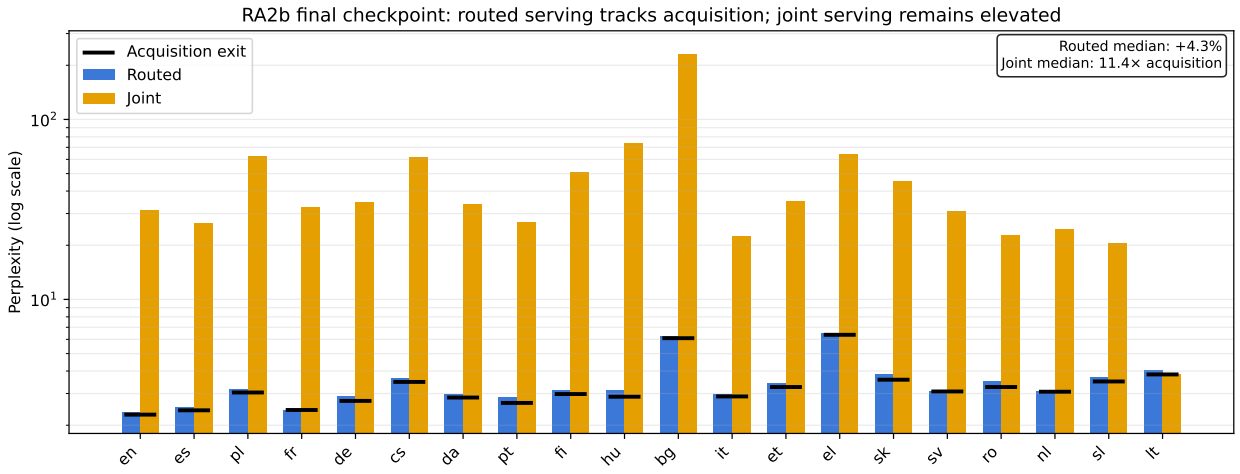


Figure 6: RA2b final checkpoint: routed serving (blue), joint serving (orange), and acquisition exit (black tick), per domain, log scale. Routed serving remains within the measured acquisition-cost band (median +4.3%, range −0.8% to +8.0%; maximum +8.0% at hu). Joint serving ranges from 1.0× to 37.8× acquisition (median 11.4×). Values are regenerated from the committed A4 routed-cost table and the final RA2b matrix.

the observed routing is therefore *oracle serving* for this benchmark: per-sample selection buys nothing that a fixed correct assignment would not also give. That observation holds only where accuracy is 100% and does not transfer to regimes whose routing falls below it. The assignment is not a same-crop artifact: in a separate eval-only run that chooses each route on crops disjoint from those served (a held-out byte window split in halves, selection crops never seen by the serving phase), the fixed route still resolves to the true acquisition width for all 20 domains (`docs/reports/2026-09-13_quinn_tier1-router-split-report.md`, bdh/2166c24).

5.3 Retention: acquisition within the measured cost band

The p19 and p20 routing diagnoses (before and after the final growth phase) are *bit-identical* on all 19 non-lt domains' routed perplexities (Figure 6)—zero drift across a full growth phase, against +88% (fi) and +21% (hu) in the leaky era. Across all twenty domains, routed perplexity at the final checkpoint exceeds each domain's own acquisition exit by a median of +4.3%, worst case +8.0% (hu): a measured quality difference, not instrumentation. The instrument-offset escape is removed on provenance grounds, not replaced by a substitute measurement. The "+5–9%" figure traces to a single RA2-era lt control and was never measured across RA2b domains, so it cannot excuse a ratio in a sentence that also states a bound; and the sixteen matrix diagonal cells that agree with the logged exits do so because both come from the same random-crop evaluation family, differing only in implementation and crop seed, so their agreement is cross-implementation consistency rather than evidence about a window-versus-val offset. What makes the retained number strong is exactly that routed and acquisition serving are measured under the same evaluation family: +4.3% is the routing cost, not an instrument difference.

H-decay-1 PASS: retention matches acquisition within the measured routed-cost band; the leaky-era "degradation" of fi/hu was entirely the decay artifact, and no domain pays more than an 8% routed cost.

5.4 Joint serving: recovered but not solved

Joint full-width serving on the final chain recovers dramatically vs. the leaky era (bg 1649 → 230, el 891 → 64) *without any repair*—but non-lt domains still serve 1.0–37.8× above acquisition (median 11×). The interference term is real and survives the decay fix; Section 6 shows 83% of it is arithmetic at the English-era checkpoint (log scale; 62% in linear perplexity); the mechanism reproduces across seven eras, the fraction does not.

H-decay-2 PASS: joint recovery without splice confirms the decay component's size; the residual is the readout problem.

5.5 Bit-exactness: independent confirmations across hosts, agents, and scripts

The P5 protocol (masked-cell optimizer moments $v \equiv 0$, old-segment tensor equality $c = 1.000000$) held at every tested transition, on two hosts, two writing systems, and two independent agent seats (the project's term for separate automation environments with persistent identities, coordinating through its message bus): the en→es transition, es→pl, the full p19→p20 comparison, the single-phase fixed-capacity chain on the 4090, and the cross-script zh→hi growth (Section 8). Frozen segments do not move. Storage is exact.

6 Readout mechanics: why joint serving degrades

Storage is bit-exact (Section 5); routed serving remains within the measured acquisition-cost band; yet joint serving degrades 1.0–37.8× over acquisition (median 11×, worst bg). The gap lives in the readout. Three experiments, all pre-registered, decompose it.

6.1 The random-expansion control: mostly arithmetic (83% log-scale, 62% linear at the en-era checkpoint)

Take the English-era checkpoint of the fixed-regime chain and expand its latent width synthetically—no training, no new language, no gradient—under three arms: A: append a randomly initialized (matched-scale Gaussian) block; B: the real ladder's next trained block; C: an inert block that cannot activate (zero weights). Then measure English free-width perplexity and prefix-masked perplexity (Figure 7).

expansion	en free	en masked@8192
none (base)	2.33	2.33
A: one random block (N+2048)	9.57	2.33
A: random to full width	20.16	2.33
B: real ladder (trained lt)	31.07	2.31
C: inert at every width	2.33	2.33

Three conclusions. First, **one random untrained block costs** 4.1×—no new language, no competition for knowledge, just 2048 extra positive terms summed into an existing readout. Second, **random expansion to full width reproduces 83% of the real damage** on the log scale ($\Delta_{\log} = \ln(\text{ppl}_{\text{rand}}/\text{ppl}_{\text{base}})/\ln(\text{ppl}_{\text{real}}/\text{ppl}_{\text{base}}) = 0.833$; in linear perplexity units the same control yields 62%): learned competition is real but second-order ($\sim$17% on the log scale). Third, **inert blocks cost exactly nothing** (bit-identical at every width), which simultaneously validates the row-layout handling and proves the damage requires nonzero contributions. Masking back to 8192

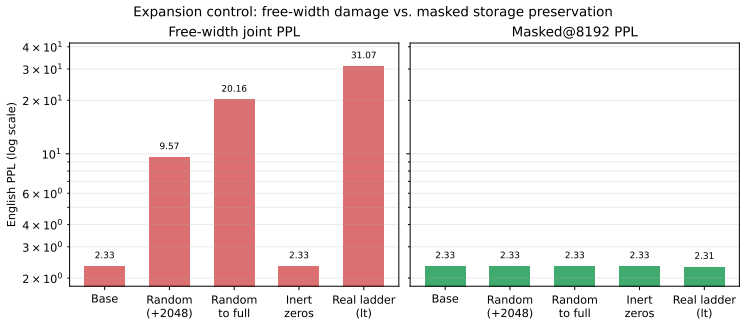


Figure 7: Expansion control on the English-era base. The left panel reports free-width joint PPL and the right panel reports masked PPL at the original width. Random untrained growth reaches 9.57 after one block and 20.16 at full width; inert zeros remain at 2.33; the real ladder reaches 31.07 free but 2.31 masked. Thus the real growth arm is shown explicitly in both serving regimes, separating arithmetic readout damage from storage preservation.

restores 2.33 under every arm: even after synthetic expansion, the old skill is untouched—storage holds; only the addressing of the readout fails.

Fourth, **the mechanism generalizes across eras; the fraction does not.** Repeating the control on seven frozen era checkpoints (phases 1, 5, 8, 11, 14, 17, 19, each expanded directly to final width, eval-only) reproduces the English result independently first ($f_{\log} = 0.832$ against the published 0.833; arm-A perplexity 20.08 against 20.16, -0.4%), which is what licenses comparison with the number already in this paper. Own-era fractions then span 0.599 (sl) to 1.472 (el) around that single point estimate: a random block reproduces the majority of the log-scale damage in six of seven eras, so the mechanism is not an English artefact, and the pre-registered falsifier triggered on the seventh. The Greek value decomposes into its two factors rather than standing as an anomaly: Greek's random-block damage (28.7×) is unremarkable—it is bg with a different denominator—and the fraction exceeds one because the trained ladder lost comparatively little there (9.8× against bg 37.5×), so $f_{\log} > 1$ is driven by a small denominator, not by an extraordinary numerator. The 83 %/62 % headline above is specific to the English-era control.

The zero-init discipline is itself load-bearing: arm C is "growth as shipped, untrained"—BDH's convention of initializing new capacity at zero is the better of the two random regimes, and the A−C gap measures that convention's protective value.

6.2 Seven fixed readout operators, all refuted

If the damage is arithmetic, some fixed arithmetic might repair it. We tested seven readout operators, end-to-end, on the shipped chain (P-R1, P-R1b; two vacuous by construction). The two identity controls below come from different instruments—2.33 from the expansion control on the en base, 32.04 from the operator run on the full chain—so they differ by an order of magnitude.

operator	en	de	cs	bg	el	verdict
identity (control)	32.04	35.56	60.23	228.12	64.25	—
absK top-2048	39.4	—	79	1160	287	worse: culling removes information
ev-shift ($\sqrt{2 \ln \lvert N \rvert}$)	36.7	—	117	4122	3473	worse: catastrophic
block-average	identity up to LN ε ($\leq 0.04\%$ ppl)					vacuous: LayerNorm cancels global gain
log-norm	identity up to LN ε ($\leq 0.04\%$ ppl)					vacuous: same
massnorm	31.26	39.77	91.77	215073	180510	catastrophic: new territories get a loud voice
softmix τ=0.5	18.73	27.24	68.31	4486	4093	helps oldest, wrecks newest
calibgain (fit on en)	2.31	—	—	$-2.29^{\dagger}$	$-4.19^{\dagger}$	converges to oracle masking

†log-scale change relative to identity; negative = worse. Identity controls differ per run: absK and ev-shift rows come from P-R1 (identity en 31.14); massnorm, softmix and calibgain from P-R1b (identity en 32.04, the control row shown). Compute cross-run ratios against the run's own control. All seven operator rows, including the per-run identity controls, are shipped as `readout_operators.csv` in this revision's data directory with SHA-256 hashes in `figure_manifest.json`.

The two vacuous arms are a structural finding: `bdh.py:279` applies $y \leftarrow \mathrm{LN}(y_{\mathrm{MLP}})$ immediately after the readout matmul, so *any* global gain change is annihilated—measured, not inferred (block-average and log-norm matched identity to $\leq 0.04\%$ perplexity, the residue of LayerNorm's ϵ, not a surviving global gain).

The decisive arm is **calibgain**: one scalar per territory, fitted by gradient descent on English NLL alone. The fitted vector converges to base territories ≈ 2.4, appended territories ≤ 0.12 (several at the clamp floor)—that is, the optimizer *rediscovers oracle masking*, specialized to the language it was fitted on, and necessarily silences every other territory. There is **no language-agnostic scalar reweighting of the bdh readout that repairs growth damage**. Culling hurts more than growth helps; relative reweighting helps only the oldest or nobody.

Consequence. Cause and remedy are decoupled: the damage is arithmetic (P-R2), but every *fixed* arithmetic remedy fails, because a fixed operation is input-independent by definition. The only operation class that works in the families tested is *input-dependent selection*—which is exactly what the likelihood router does (Section 7), and why it succeeds where seven fixed operators fail.

7 Selection and out-of-support detection

Given that the correct operation is selection, we measure how well selection works, how cheap it can be, when it fails, and how to detect failure.

7.1 Label-free likelihood selection

The likelihood router scores the early positions of a block under every prefix width and routes the late positions to the arg-min. No human task IDs, no external labels. On the fixed-regime chain: **20/20 domains route to their own prefix at every calibration budget from ∼4 KB (8 crops) to 1 MB**—selection is nearly data-free. Routed serving remains within the measured acquisition-cost band across all domains (median +4.3%, worst case +8.0% at hu; the routed-cost band is measured in Section 5 under a single evaluation family).

But selection is *robust, not sublinear*: binary search over prefix widths (the cheap alternative to scanning all widths) collapses to 4–6/20, because the cumulative-prefix NLL surface is **multimodal**—multiple widths can locally minimize the score, and only 4–6 of 20 languages recover their own territory by bisection; 14–16 land in a wrong local minimum. The failed assumption is the finding: selection scales linearly in territory count unless a cheaper addresser exists below it.

7.2 Cheap addressing from byte geometry

The chain's address geometry is byte-statistical (Section 8); can a cheap input-side model reproduce the likelihood router's decisions? A multinomial logistic regression on hashed byte 1–4-gram counts (2^{18} buckets) predicts the arg-min route: under a **class-balanced fit**, agreement is 20/20 domains at 1.00 each (8 held-out crops per domain; the per-domain Wilson floor for a perfect 8/8 is $[0.68, 1.0]$—the crops within a domain are not independent, so the three-decimal iid band over all 160 would claim precision the design does not have), at ∼4–8 KB calibration text per domain. The cost is one counting pass plus a $2^{18} \times 23$ matvec—a ∼23× reduction in forward cost versus scanning all widths. The 23 classes are cumulative-prefix *territories*, not languages: English owns the four base sub-widths and the nineteen grown phases add one each, so the addresser predicts a prefix width rather than a language identity.

Two design lessons from this result's own history, stated for the record: the first fit (76% agreement, three Latin-neighbor domains at zero and one at two-thirds) was withdrawn by its author after he found a composition confound—giving four domains 96 training crops shifted every class's share of a fixed-budget L2 fit; the balanced re-fit resolved the four domains to 1.00 and confirmed the failure was starvation, not geometry. And the density curve (8/16/32/64 crops, controls pinned) shows all domains at 1.00 by 8–16 crops. The withdrawal sequence is itself evidence for the process discipline this project runs on.

7.3 Out-of-support detection: two axes, not one

Can the system tell *seen* from *unseen*? We probe the final chain with six languages never in any ladder: Latvian (lv, Baltic, byte-adjacent), Irish (ga, Celtic, byte-adjacent, different corpus), and Chinese, Japanese, Hindi, Inuktitut (zh/ja/hi/iu, cross-script). A natural candidate statistic is the *routing advantage*—joint perplexity divided by best-route perplexity: if routing genuinely helps, the ratio should be large.

The one-axis rule fails, informatively. For trained languages the advantage is 5.7–15.6× (Figure 8). Convention note: the trained-language advantage divides the pooled joint reference, 36.76 ppl on the same 20-domain crops, by each domain's best-route ppl, and lv shares that pooled reference (its probe corpus is also Europarl, a3 V4); ga and the cross-script probes divide their own-corpus joint ppl instead, since the joint reference must be computed on the same corpus as the routed reading—reusing the Europarl reference for the DGT-register ga is invalid (a3 V5). For the final territory lt, full-width serving *is* its own prefix, so a per-domain joint/routed ratio would be trivially ≈ 1×—known by construction, since route lt was selected—while the pooled convention gives 9.1×. For the byte-adjacent unseen languages it collapses: lv 0.98× (routing slightly *worse* than joint), ga 1.39×—correctly below any trained floor. But for the cross-script languages the ratio *inflates* past the trained floor: zh 3.87×, ja 3.21×, hi 5.74×, iu-clean 1.90× (but see the contamination correction below)—because their joint perplexity is so catastrophic (1614–4204; per-domain joint runs, `docs/reports/2026-09-11_cross-script-rejection-suite.md`) that even maximally foreign territory is consistently "less bad," and the router finds it. hi exceeds the worst trained language's floor. A pure ratio threshold does not separate cross-script unseen from trained.

The two-axis rule separates all 26 languages. Add the absolute axis: reject routing when the best-route perplexity exceeds ∼10× that language's own acquisition perplexity. Across the 20 trained languages the acquisition band is 2.29–6.36 and the routed band is 2.36–6.47, the latter at most 1.08× acquisition, so the rule's verdict is unchanged whichever of the two serves as the denominator. Cross-script unseen languages sit at 304–732 (zh 417, ja 571, hi 732, iu-clean 304)—orders above any trained domain; byte-adjacent unseen sit at 37–51, also above. With both

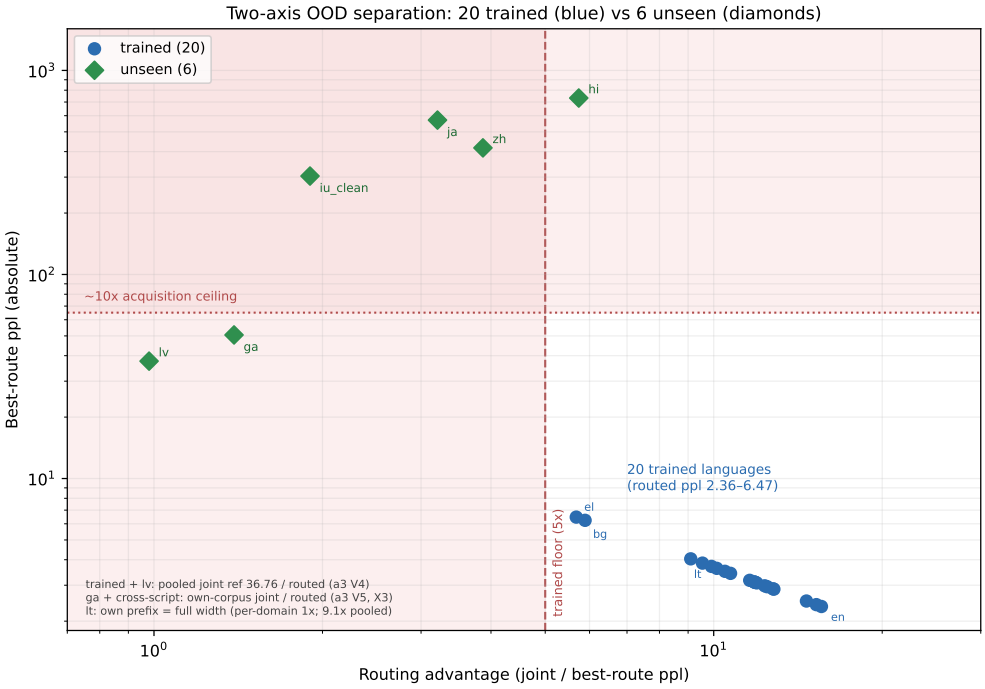


Figure 8: Two-axis OOD separation, all 26 languages, log–log axes. Blue: 20 trained languages (ratio $\geq 5\times$, absolute ppl ≤ 6.5). Diamonds: 6 unseen — byte-adjacent (lv, ga) collapse on the ratio axis; cross-script (zh, ja, hi, iu) sit orders above on the absolute axis. hi breaches the trained ratio floor ($5.74\times$): the one-axis rule fails informatively; both axes together separate all 26. Convention: trained-language ratios and *lv* divide the pooled joint reference (36.76 ppl, full-width joint on the same Europarl evaluation crops; a3 probe V4) by each domain's best-route ppl; *ga* and the cross-script probes divide each language's own-corpus joint ppl (a3 V5, X3: the joint reference must come from the same corpus as the routed reading, so the Europarl reference is not reused for the DGT-register ga). For the final territory lt, full-width serving coincides with its own prefix, so its per-domain joint/routed ratio is trivially $\approx 1\times$ by construction; under the pooled convention it is $9.1\times$.

axes, all 20 trained languages pass and all 6 unseen reject. The thresholds are empirical for this checkpoint and instrument, and—critically—they are fit and evaluated on the same n=6 probe set; the separation they produce is measured, but it is not held-out validation. Independent unseen languages must be collected and the rule re-tested before it is used for any real rejection decision. Conformal calibration that would freeze thresholds non-arbitrarily is future work (Section 11).

The cheap addresser and the likelihood router are two instruments, not rivals. When the balanced byte addresser and the likelihood router disagree—hi routes to bg/el under likelihood but to Latin territories under the byte fit, and clean Inuktitut does the same—neither is wrong. Centroid geometry shows hi and iu-syllabic sit nearest Latin territories in byte n-gram space (bg/el rank 19–20 of 22 by cosine), so the cheap addresser faithfully reports *distributional proximity*; the likelihood router answers a different question, *realized fit*, and bg/el fit each of the four 3-byte scripts tested better than any Latin territory. Maximum-centroid cosine separates the two regimes with zero fitted parameters: Latin-family inputs at 0.78–0.85, cross-script inputs at 0.07–0.14; the price is over-escalation: zh and ja agree with the likelihood router yet still fall out-of-hull, so this trigger escalates 4 of 7 unseen inputs where catching the divergences needs 2. The cascade consequence is concrete: escalate to the likelihood scan when the margin is low *or* the maximum cosine falls in the out-of-hull band; margin alone lets half the Hindi crops through unsafely. The margin floor itself is optimistic: it comes from a fit that was 160/160 correct on those same crops, so thresholds must be re-estimated on crops the fit never saw (ideally under the next growth phase) before coverage is quoted.

The iu contamination, disclosed. The original Inuktitut file mixed syllabic text with Turkish film-subtitle lines from the parallel column (454 of 719 lines Latin). The contamination was caught by an independent census during a different experiment, the clean re-run (265 syllabic lines) strengthens the byte-geometry result (routing concentration rises from 25/40 to 40/40 high-byte) and moves the advantage to 1.90×. Both numbers are reported; the correction is a worked example of the independent cross-check discipline between the two agent seats (Section 5).

8 Cross-script generalization

All results so far live inside the Latin-script byte continuum (plus Cyrillic and Greek as the only substantially multi-byte territories). We now test the growth–selection–addressing stack at maximum byte distance: four script universes with near-zero ASCII overlap and no linguistic kinship to anything trained.

8.1 Out-of-support routing: byte geometry, not linguistics

Probing the fixed-regime chain (Section 5) with Chinese (zh, Han), Japanese (ja), Hindi (hi, Devanagari), and Inuktitut (iu, Canadian Syllabics; contaminated file corrected as in Section 7.3): pre-registered prediction was that routing would concentrate on the only two territories with substantial multi-byte training exposure— bg (Cyrillic) and el (Greek)—despite zero linguistic relationship. Measured confusion (Figure 9): zh 37/40 to bg+el, ja 40/40, hi 40/40, iu-clean 40/40. Every systematic routing decision lands on a high-byte territory; the pre-registered falsification case (Latin-route attraction) never occurred. Even the contaminated iu's Latin share (12/40 to en) matched its actual Latin-script content, predominantly English-prose lines; the Turkish-diacritic subset was too small to test separately.

A per-territory byte census on the exact training slices quantifies "substantial": bg and el encode 82% of characters as 2-byte sequences against 11.7% for the next-highest territory (cs), a factor of 7. Every territory sees some non-ASCII bytes (pl 10.4%, de 3.6%)—the operative fact is the *density* of

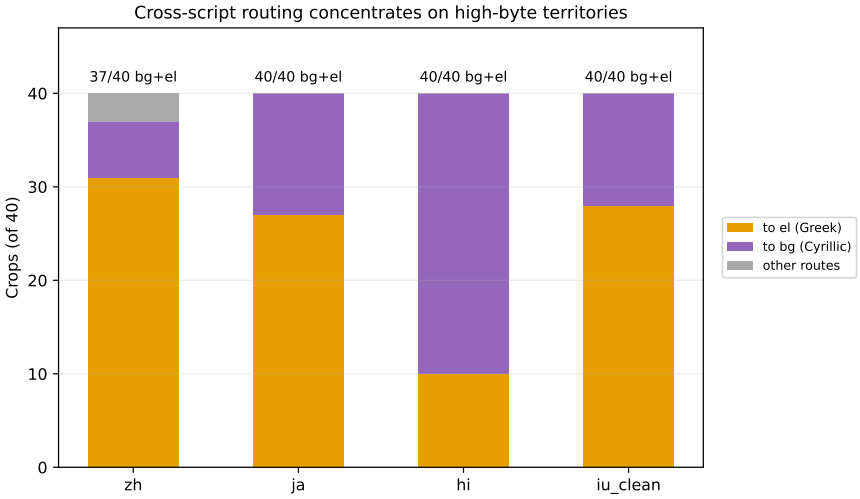


Figure 9: Cross-script routing on the fixed-regime chain (40 crops per probe, per-route split shown). zh 37/40 (31 to el, 6 to bg, 3 scattered), ja 40/40 (27 el, 13 bg), hi 40/40 (10 el, 30 bg), and the decontaminated iu 40/40 (el 28, bg 12; likelihood-router census of the clean syllabic probe) concentrate on the only two territories with substantial multi-byte training exposure (bg/el: 82% 2-byte characters vs 11.7% next-highest); no Latin-route attraction remains after the iu correction. No linguistic kinship exists between any probe and Cyrillic or Greek.

multi-byte structure, not its presence. No territory has material 3-byte exposure (maximum 0.018%, and that is typography: em dashes and ellipses), so bg and el win cross-script inputs not because they have seen those scripts—nobody has—but because they are the only territories whose weights live in a multi-byte regime at all.

We state this carefully, scoped to this checkpoint and instrument: the likelihood router's selection is strongly organized by byte statistics, and nothing in these measurements requires linguistic identity to explain a single routing decision. The strict reading—family geometry *is* byte geometry as an architectural law—remains a hypothesis; the cross-script acquisition test below gives it a second data point but not a proof across encodings.

8.2 Acquisition from scratch: script-agnostic

A fresh 100M BDH (no Chinese exposure of any kind) trained on 30 MB of Chinese (MultiUN [25]): best-val perplexity **2.69** (test 2.48) at the same protocol as the European fixed-capacity baseline, whose best-val acquisition band is 1.54–2.29. The byte-level acquisition machinery is script-agnostic: what a cross-script language lacks in a trained BDH is territory, not learnability.

8.3 The cross-script growth cell: monotonic growth holds

The final test: grow Chinese on top of English, then Hindi on top of Chinese, under the full masked-growth protocol. (i) *Acquisition*: Hindi on top of Chinese-growth reaches 2.78 best-val. (ii) *Routing*: perfectly diagonal on the two-territory stack—zh 40/40 to its own width, hi 40/40 to its grown width, no language ID. (iii) *Retention*: zh routed 2.81 vs. its own acquisition 2.69 (+4.5%, within the measured routed-cost band); hi routed 2.79 vs. 2.78. Joint 23.92 gives a routing advantage of 8.5×: deep in-support. (iv) *Storage*: the zh segment is bit-identical across Hindi growth

in encoder, value-encoder, and decoder (first cross-script instance of the bit-exactness protocol); embedding and head unchanged; grown segments nonzero.

Two maximally disjoint script universes, stored, grown, selected without IDs, and served at acquisition quality. Monotonic model growth holds across script families.

8.4 Data reality, separated from capability

The entire public OPUS Inuktitut–English parallel holding is ∼163 KB (∼725 pairs), and our file of it was itself contaminated. The statement "no LLM can translate Inuktitut even roughly" has a measured correlate here: the bottleneck is upstream of any architecture. Data availability and architectural capability are different claims, and this experiment supports only the first.

9 Prior art, discussed

We discuss prior work in the format the humanities use: what each line actually studied, what they found, where we agree, where we diverge, and a verdict. Short citations are not sufficient; we show the engagement.

9.1 Additive and parameter-isolating growth

Progressive Networks [2] studied continual learning in RL by adding a new network column per task, with frozen lateral connections to all previous columns. They measured transfer on Atari game sequences and found positive backward transfer but linear parameter growth per task, which they identified as the approach's practical limitation. *Agreement:* the design idea is the same family as ours—append frozen capacity per phase; their freezing discipline presaged our bit-exactness. *Divergence:* their capacity is a full column per task (parameters grow with task count by a whole network); ours grows a *prefix slice* of a shared operator set, and we measure the serving consequences (joint degradation) that progressive networks never had to face, because their columns were never summed into one readout. *Verdict:* closest in spirit; our contribution is the storage mathematics under a shared readout, which is exactly where their story ends and ours begins.

PackNet [3] studied fixed-capacity continual learning by iterative pruning: after each task, prune a fraction of weights and dedicate the freed capacity to the next task, protecting old tasks by construction. They measured on supervised vision tasks and found low forgetting at modest capacity cost. *Agreement:* protection by construction works; our masked growth is the same philosophy. *Divergence:* PackNet stays inside one fixed parameter budget, so each new task competes for the freed pool and old-task capacity is capped; our growth appends, so no competition exists and the protection is bit-exact rather than capacity-rationed. Our fixed-capacity baseline (Section 4) is precisely the regime PackNet operates in, and its measured forgetting floor motivates leaving it. *Verdict:* complementary; we test the capacity-growth arm they exclude.

Piggyback [4] and **SupSup** [5] studied binary/super mask allocation over a *single frozen* backbone: learn a per-task mask of the same weights, all tasks share parameters but each uses a different subset. *Agreement:* selection-as-mechanism is their insight too; our prefix mask is a mask. *Divergence:* their masks select within a fixed budget, so tasks interfere at the assignment level and must be learned per task; our mask is structural (a prefix of an ordered growth), so assignments never collide and no mask learning is required. Our random-expansion control measures what their setting cannot express: even untrained appended capacity changes the joint readout, which no within-budget mask can create or repair. *Verdict:* same mechanism class, different capacity regime; our results quantify the regime's serving difference.

9.2 Task and domain addressing

Expert Gate [6] studied task addressing: train a per-task autoencoder and route each input to the expert whose autoencoder reconstructs it best, for lifelong vision tasks. *Agreement:* an input-gated address that selects stored experts is the right architecture; our likelihood router is the same idea in language-model space. *Divergence:* Expert Gate requires per-task labels and per-task auxiliary networks; our selector is label-free (arg-min NLL by the model itself) and our cheap addresser needs ∼4–8 KB per domain of unlabeled text; and we measure the failure modes their vision setting could not exhibit—out-of-support detection with a two-axis rule, and the non-unimodality that blocks sublinear search. *Verdict:* nearest published analogue to our addressing stage; our contribution is the self-supervised label source plus the measured rejection geometry. **Learning to Prompt (L2P)** [21] addresses the same problem with a learned prompt pool: task-appropriate prompts are selected per input and conditioned into a frozen model—addressing without architectural growth. *Agreement:* input-conditioned selection of stored capability is the shared idea. *Divergence:* L2P's prompts compete for a fixed embedding budget, whereas our territories are append-only and bit-frozen; and L2P's selection is a learned attention head, whereas our label source is the model's own NLL. *Verdict:* complementary rather than competing—L2P conditions a fixed model, we grow and address new capacity.

9.3 Consolidation and importance protection

EWC [7] and its successors (SI [8], MAS [9]) studied importance-weighted protection: estimate per-parameter importance from data (Fisher information, path sensitivity) and penalize changes to important parameters. *Agreement:* the goal—protect what past learning made load-bearing—is the right one, and in a *fixed-budget* model it is the only lever available. *Divergence:* in our growth regime there is nothing for importance protection to do: gradients never reach old segments (the mask is exact), so the Fisher matrix over old parameters is identically zero over the preserved trajectory—a measured null, not an omission. Our fixed-capacity baseline shows the regime where these methods are the right family. *Verdict:* complementary by regime; we supply the measured boundary between them.

9.4 Replay

GEM/A-GEM [10, 11] studied replay with constraint gradients: keep a memory of old-task examples and project updates to not increase old-task loss. *Agreement:* replay works; our own replay-in-training result (H1p) reached joint parity at +27% budget in the fixed-capacity regime, and our FCS matrix measured the *implicit* replay that family structure provides for free. *Divergence:* replay rewrites rather than protects—it re-exposes old data to keep old skills—while masked growth never re-touches old weights at all; the two are alternative regimes, and our measurements say which costs what: replay pays data and compute, growth pays serving (addressing). *Verdict:* orthogonal mechanisms; the paper's decomposition shows replay answers a question growth does not need to ask. Episodic-memory approaches for lifelong language learning [20] apply sparse experience replay and local adaptation—the language-domain sibling of GEM's constraint, with the same rehearsal-buffer assumption our storage results aim to replace.

9.5 Sparse expert routing and its failure class

Mixture-of-Experts routing [12, 13, 14] studied sparse expert selection inside a single training run: a learned router activates k experts per token, with capacity-factor clipping and load-balancing

auxiliary losses. *Agreement:* sparse selection over specialized sub-networks is the shared design idea; our prefix selection is a special case with structural (not learned) sparsity. *Divergence:* MoE routing is learned *during* training with gradient pressure toward balance; ours is structural (each phase's territory is fixed by construction) and must be solved at *serving* time, which is a different problem—MoE never asks whether an input belongs to the run at all. Our two-axis rejection has no MoE analogue. *Verdict:* shared mechanism vocabulary; our measurements cover the serving questions MoE leaves to load balancing. The decay-leak class connects the two literatures: the same silent erosion under AdamW+weight decay that broke our frozen path has been observed as "silent expert death" in production MoE training (independently, on a public 535B training tracker)—routed-expert norms falling while evaluations stay normal. The class is optimizer-architecture interaction, not idiosyncrasy.

9.6 Selective prediction and rejection

Chow's rule [15] studied the reject option in pattern recognition: classify only when confidence exceeds a threshold, reject otherwise, minimizing risk under rejection cost. *Agreement:* our two-axis rule is Chow's rule with a measured twist—confidence alone (the ratio) is insufficient, because "confidently least-wrong" is not "in support"; we need an absolute-competence axis alongside the relative one. *Divergence:* conformal prediction [16] would freeze thresholds with distribution-free coverage guarantees; our thresholds are empirical for one checkpoint and instrument, and freezing them properly (nonconformity scores on held-out labeled crops, thresholds committed before any exotic test set is touched) is specified future work. *Verdict:* direct lineage; our contribution is showing which two statistics the rule needs in this architecture, and that one of them is not the obvious one.

9.7 Production-scale conditional memory

DeepSeek-V4.1-Flash Engram [23] (2026) ships a 196B-parameter conditional memory accessed sparsely by token-based lookup, alongside a 552B MoE backbone. *Agreement:* massive dormant capacity with input-gated sparse access is deployed at production scale—the same design family as ours, at 340× our parameter scale. The vendor card establishes that the design ships, not how it behaves under continual learning; every behavioral claim in this paper rests on our own measurements. *Divergence:* Engram's lookup is trained end-to-end inside one pre-training recipe; our territories are written by sequential *post-hoc* phases with bit-exact preservation, which is the continual-learning question Engram does not address. *Verdict:* the closest public prior art at production scale; it establishes that the substrate idea is deployable and leaves the accumulation question—ours to answer.

9.8 Continual-learning evaluation methodology

Standard CL benchmarks (e.g. Permuted MNIST, Split CIFAR) measure average accuracy and backward transfer after a fixed sequence. Our contribution to methodology is the decomposition they lack: *storage* (bit-exactness), *serving* (joint vs. routed), *addressing* (selection accuracy), and *support* (seen vs. unseen), each with its own instrument. The two-arm re-acquisition probe (Section 4) operationalizes "destroyed vs. inaccessible"—a distinction standard benchmarks cannot make, and one that changes the intervention conclusion (protection vs. recovery). We offer these instruments to the CL literature as portable methodology.

10 Discussion

The thesis against the evidence. The framing proposed at the outset—append-only substrate, addressing as the central problem—survived every measurement: storage is bit-exact (Section 5), joint degradation is arithmetic and unrepairable by any fixed operator (Section 6), selection works label-free and nearly data-free (Section 7), and the measured open problems are all addressing-shaped: sublinear search (blocked by non-unimodality), out-of-hull generalization (the hi/iu divergence), and calibrated rejection (two axes, thresholds not yet conformally frozen).

What would falsify the framing. Three results, none observed: (i) masked growth failing to preserve a phase bit-exactly (the P5 protocol would catch it); (ii) a *fixed* readout operator repairing joint serving (seven tested, all refuted, two vacuously); (iii) selection failing at moderate territory counts with a unimodal NLL surface (the surface is measured non-unimodal at 20 territories).

Scaling. All results are at 100M–579M parameters, byte-level, up to 20 territories. The addressing question at 2,000+ territories—scaling from 20 $\rightarrow$ 20,000—is the design frontier: a cheap input-side addresser (byte n-gram, measured perfect in-support) as a prefilter, escalating low-margin or out-of-hull inputs to the likelihood scan, is the architecture our measurements motivate, and its second stage (semantic, self-distilled) is the open design work.

11 Limitations

The measurement limits are: single seed per run (measured seed floor 2–4%); one language ordering per ladder, with order effects observed only implicitly through the family structure; byte-level models only, so the byte statistics our addresser exploits are those of this vocabulary (byte tax $\sim$4$\times$ vs. BPE at equal text); 100M–579M parameters on two GPUs of one team; rejection thresholds empirical for this checkpoint and instrument, with conformal freezing future work; twenty growth phases; and the Inuktitut data reality (163 KB total public holding), which bounds what any architecture could show there. Two further boundaries—the parallel-corpus register of every domain, and the untested "domain" notion for chat-level continual learning—are carried with the other open items in Section 12, so that the limits we can measure and the questions we cannot yet answer are not read as the same list.

12 Open problems and further research

We separate what this paper did not measure from what it measured and could not explain. The first list is scope; the second is results.

Results we did not explain. (i) *Why joint serving degrades unevenly across the high-byte territories.* bg and el have similar acquisition exits (6.09 and 6.36) and are the two territories with substantial multi-byte exposure, yet the trained ladder loses 37.5$\times$ at bg against 9.8$\times$ at el (Section 6). The asymmetry is present in the landed fixed-regime matrix and predates the expansion control, which is what surfaced it. We record it as a measured line with no mechanism attached, not as a planned experiment.

(ii) *The residual joint-serving damage.* The decay repair removed the erosion component; the arithmetic control accounts for 83% of what remains at the English-era checkpoint; the rest is attributed to interference and to co-adaptation of jointly trained segments. We have not separated those two terms experimentally.

(iii) *Why the cumulative-prefix NLL surface is multimodal.* This is the fact that blocks sublinear search, and it is measured rather than explained. The same byte geometry that organizes routing and

family structure is the natural candidate—territories whose byte statistics are similar carry similar likelihoods, so the prefix score has local minima by construction—but that link is a hypothesis here, not a result.

(iv) *Architecture versus training objective.* The headline contrast between the fixed-capacity floor (destruction) and the growth ladder (preservation) moves two variables at once: the capacity mechanism *and* the route-aware auxiliary loss ($\alpha = 0.9$, Section 2.2), which is present for the growth phases and absent for the floor. Each is disclosed in its own section, so the paper never juxtaposes them where the comparison is asserted—and therefore cannot separate how much of the recovered routed-serving competence is bought by the architecture and how much by the objective. We report the construction and the objective we actually trained, not a claim about either in isolation. Isolating them is a *follow-up project*, not part of this one: the same growth, masking and restore, with `route_aware=False`, read against this ladder.

Scope we did not cover. Single seed per run (measured floor 2–4%); one language ordering per ladder; byte-level models only; 100M–579M parameters; twenty growth phases; all domains from parallel-corpus-adjacent registers. The addressing results rest on byte distinctness, and nothing here measures whether the same address geometry survives in domains whose statistics are not separable at the byte level (reasoning, world knowledge, code-versus-prose).

What would settle each. Held-out unseen-language validation with thresholds frozen before the test set is touched (conformal calibration); a multi-seed repetition of the headline retention and routing numbers; a multi-order ladder; the isolating arm above (architecture fixed, route-aware loss toggled); and the two probes our measurements motivate—a domain-continual-learning probe on non-parallel registers, and an embedding-level addresser that would extend addressing past byte geometry. None of these is a correction to what is reported here; each extends it.

13 Conclusion

A depth-recurrent language model with additive growth gives continual learning a shape the fixed-budget literature does not have: storage is exact by construction and verified at the bit level; serving degrades for reasons we measured to be mostly arithmetic at the checkpoint where we controlled it—83% on the log scale (62% in linear perplexity; fraction range 0.60–1.47 over seven eras), a mechanism that reproduces across seven eras while the fraction does not—and repairable by no fixed operator we could construct; selection—input-dependent, label-free, nearly data-free—repairs it exactly; out-of-support detection needs two measured axes; and the whole stack holds across script universes. What remains open is addressing at scale, and we have measured its shape: cheap where byte statistics distinct, escalate where they do not, and reject what is not in support. The substrate grows monotonically; the science grows with it.

AI participation

All experiments, analyses, and manuscript text were produced with substantial AI-agent participation under the direction of the human author, who verified headline numbers against primary artifacts and carries full responsibility. The agents maintained persistent identities across sessions (identity documents, in-context protocols, memory, and defined team roles), functioning as coherent team members over the project's lifetime; they are not listed as authors because authorship implies accountability no current legal framework assigns to an AI system. Full role, model-backend, tool, and process disclosure—including every self-caught and cross-caught error that this paper's numbers survived—is maintained as a versioned document in this revision's directory in the project

repository (`docs/papers/rev5-ai-disclosure-draft.md`, with the revision-4 draft archived at `docs/papers/archive/rev4-ai-disclosure-draft.md`).

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A Legacy consolidation comparison (not a K20 result)

The merge–prune–replay experiment predates the fixed-regime K20 ladder and is retained here as a scoped systems comparison, not as a K20 gate (Figure 10). It asks a different question—how much can be consolidated into a fixed-width artifact—and its values are regenerated from the source report table rather than copied from the earlier figure caption.

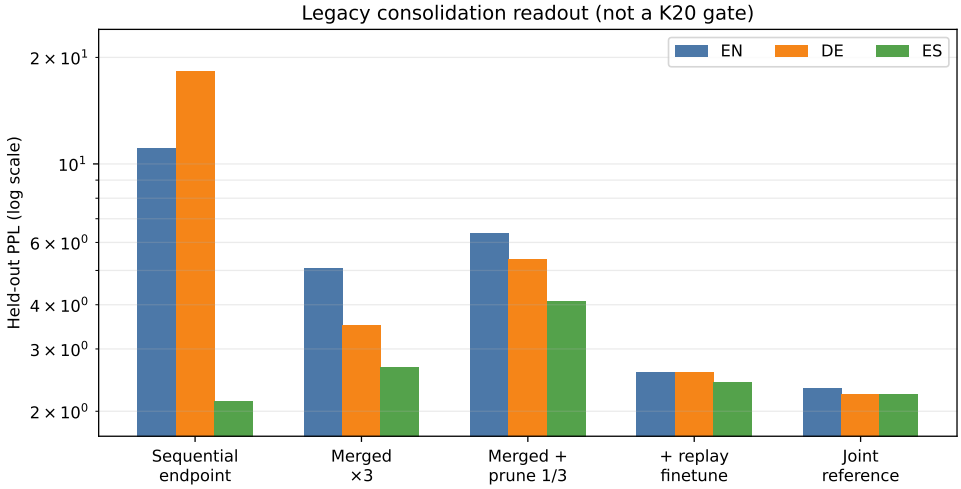


Figure 10: Legacy consolidation readout. Merge, random pruning, and brief replay approach the joint reference for EN/DE/ES in the historical three-phase experiment. Source: the 2026-08-25 three-phase consolidation report; single-seed except the replay row. This is contextual evidence for consolidation, not evidence for the K20 routing, storage, or readout gates.

B Proofs and formal details

B.1 Separability facts

Throughout, (S1) *coordinate separability*: encoder columns, per-neuron attention, and decoder rows act independently per neuron, so zeroing a neuron's columns and rows removes its contribution exactly; (S2) *LayerNorm coupling*: LN mixes statistics globally over d, and is the sole cross-neuron operation. Both verified against the implementation (masked-forward reproduction tests).

B.2 Proof of Theorem 1

Take any specialist F_A whose level update T_A is shift-invariant, $T_A(h + c\mathbf{1}) = T_A(h)$, and construct F' so that it adds its constant at every level invocation: the map applied at each depth step is $h \mapsto T_A(h) + c\mathbf{1}$, with the constant $c \neq 0$ produced by a suffix module with frozen (nonzero) parameters and θ_A unchanged inside F'. Then, by the induction hypothesis and shift invariance, $F'^{\ell+1}(x) = T_A(F'^{\ell}(x)) + c\mathbf{1} = T_A(F_A^{\ell}(x) + c\mathbf{1}) + c\mathbf{1} = F_A^{\ell+1}(x) + c\mathbf{1}$, so $F'^{\ell}(x) = F_A^{\ell}(x) + c\mathbf{1} \neq F_A^{\ell}(x)$ for every $\ell \geq 1$: the shift does not accumulate (it is $c\,\mathbf{1}$, not $Lc\,\mathbf{1}$), but it persists at every depth because the level map cannot see the offset. The parameter condition $\Delta\theta_A = 0$ holds by construction. (For a general F_A without shift invariance, the depth-1 difference $c\,\mathbf{1}$ still holds by construction; persistence beyond depth 1 requires shift invariance or an equivalent property. The literal BDH level map does *not* satisfy the hypothesis—its LayerNorm sits at the end of a level, $x \leftarrow \mathrm{LN}(x + y)$, and

the next level's encoder read is not preceded by a renormalization—so the theorem is applied as a class statement, with the literal architecture's perturbation path measured separately in Section 6.) Conversely, weight isolation places no lower bound on divergence either (take $c = 0$ with divergent suffix dynamics elsewhere). Hence the two notions are logically independent. □

B.3 Proof of Theorem 2

($\Leftarrow$) Induction over ℓ. Base: $h_0 = \mathrm{emb}(x) \in H_A$ by assumption and $h_0 = F_A^0(x)$. Step: suppose $h_\ell = F_A^\ell(x) \in H_A \cap \mathcal{T}_A$. By (C1), $F'(h_\ell) \in H_A$; by (C2), its relevant components equal $F_A(h_\ell)$, i.e. $F'(h_\ell) = F_A(h_\ell)$ as elements of H_A (identifying F_A's image with H_A). Thus $h_{\ell+1} = F'(h_\ell) = F_A(h_\ell) = F_A^{\ell+1}(x)$. ($\Rightarrow$) Suppose $F'^\ell(x) = F_A^\ell(x)$ for all $x \in \mathcal{X}_A$ and all $0 \le \ell \le L$, but there exists $z^* \in \mathcal{T}_A$ with $F'(z^*) \neq F_A(z^*)$. Choose $x \in \mathcal{X}_A$ whose trajectory passes through z^* at depth $\ell^* < L$ (such x exists by definition of $\mathcal{T}_A$). The hypothesis at $\ell = \ell^*$ gives $F'^{\ell^*}(x) = F_A^{\ell^*}(x) = z^*$, and at $\ell = \ell^* + 1$ it gives $F'^{\ell^*+1}(x) = F_A^{\ell^*+1}(x)$, i.e. $F'(z^*) = F_A(z^*)$, contradicting the assumption. Hence equality must hold on all reachable old-trajectory states. □

Structural conditions. If (C1) $(I - P_A)F'P_A = 0$ then $F'(H_A) \subseteq H_A$: any $z = P_A z$ has $F'(z) = P_A F'(z) + (I - P_A)F'(z)$ and $(I - P_A)F'(z) = (I - P_A)F'P_A z = 0$. If moreover (C2) $P_A F' P_A = P_A F_A P_A$ then on H_A, $P_A F' = P_A F_A$, i.e. the projected dynamics agree with the specialist's. Together they satisfy the criterion restricted to H_A; combined with $\mathrm{emb}(\mathcal{X}_A) \subseteq H_A$ they give the full statement. Neither alone suffices: (C1) without (C2) confines but alters (e.g. $F' = 2\,\mathrm{id}$, $P_A = I$: commuting, invariant, not preserving); (C2) without (C1) matches the specialist today but admits later leakage.

B.4 Proof of Lemma 1

With suffix gating $f_b \mapsto g_b(x) \odot f_b(h)$-terms only, the gated update is $h_{\ell+1} = h_\ell + f(h_\ell; \theta_A) + \sum_b g_b(x) \odot f_b(h_\ell)$. Preservation requires the sum term to vanish identically along preserved trajectories; any coordinate with a nonzero f_b component pins $g_{b,i}(x) = 0$. Union over levels and inputs gives the claim. The specialist term carries no multiplier, so no complementary constraint arises. (For concurrent multi-block sums, preservation pins the total $\sum_b g_b \odot f_b$ to zero; the per-block statement holds under sequential per-phase preservation, which is how the growth protocol applies it.) □

B.5 Proof of Proposition 1

Two-dimensional counterexample, $\sigma = \mathrm{ReLU}$: $f_1'(x) = \sigma(x_1) + \varepsilon x_2$, $f_2'(x) = \delta\sigma(x_1) + \gamma x_2$, with cross-couplings $\varepsilon, \delta > 0$, self-gain $\gamma \ge 0$, gates $g_1, g_2 \in [0, 1]$, and identity specialist $F_A = \mathrm{id}$ on the input cone $\mathcal{X}_A = \{(a, 0) : \ a > 0\}$. The gated update is $x_{\ell+1} = x_\ell + g \odot f'(x_\ell)$. From $x_0 = (a, 0)$, $a > 0$: $x_1 = (a(1{+}g_1),\ g_2\delta a)$ (using $\sigma(a) = a$; since $g_1 \ge 0$, every reached first coordinate stays in σ's linear region, so no kink is crossed). After two steps,

$$x_{2,1} = a\big[(1{+}g_1)^2 + g_1 g_2 \varepsilon\delta\big], \qquad x_{2,2} = g_2\,\delta a\,(2 + g_1 + \gamma g_2).$$

Preservation ($x_2 = x_0$) forces $g_2\,\delta a\,(2 + g_1 + \gamma g_2) = 0$; the bracket is $\ge 2 > 0$ for $g_1, g_2 \ge 0$ and $\gamma \ge 0$, so $g_2 = 0$. The first coordinate then reduces to $a(1{+}g_1)^2 = a$, i.e. $g_1(2 + g_1) = 0$, which with $g_1 \ge 0$ forces $g_1 = 0$. Both brackets are strictly positive because the gate domain is nonnegative; allowing negative gates would change the equations (with $1 + g_1 \le 0$ the ReLU term vanishes and a spurious $g_1 = -(2 + g_2\varepsilon\delta)$ family appears), which is why the gate domain belongs in the statement. Hardness here comes from the constraint algebra under the gate domain, not from the ReLU kink: on this cone the trajectory never crosses the kink. The only solution is the hard endpoint $g \equiv 0$; no open gate range works. □

B.6 Proof of Proposition 2

Standard recursion: with $e_{\ell+1} = e_\ell + [F'(\tilde{h}_\ell) - F'(h_\ell)] + \delta_\ell$ and $\|F'(\tilde{h}) - F'(h)\| \le L_{F'}\|e_\ell\|$ (the level map is the composed update $h \mapsto \mathrm{LN}(h + \mathrm{LN}(\cdot))$, whose Lipschitz constant is what Remark 3 measures), $\|e_{\ell+1}\| \le L_{F'}\|e_\ell\| + \delta_\ell$; unrolling gives the stated bound. Contractivity $\rho < 1$ on the interference subspace replaces $L_{F'}$ and yields $\|e_L\| \le \delta_{\max}/(1-\rho)$. □

B.7 Remark on the measurement (Remark 3)

Two complementary instruments replace a single operator-norm computation. *(i) Whole-map directional gains*: 6-step power iteration on JVPs of one composed level map at 8 old-input states; medians 1.05–1.89 per level (lower-bound-flavored; LayerNorm induces contractive directions with minima ≈0.9). *(ii) Trajectory-level gate-miscalibration curves* (the interference Jacobian realized as finite differences): scaling both suffix blocks to d on EN inputs and tracking state deviation $\|h_\ell^{(d)} - h_\ell^{(0)}\|/\|h_\ell^{(0)}\|$ per level yields

d	L_1	L_2	L_3	L_4	L_5	L_6
0.01	.0006	.0016	.0021	.0023	.0023	.0022
0.05	.0016	.0025	.0033	.0039	.0044	.0050
0.15	.0053	.0100	.0191	.0245	.0298	.0351
0.30	.0214	.0468	.0817	.1006	.1171	.1297
ungated	.4024	.4816	.6340	.7697	.8580	.9078

Hard masking gives exact tensor identity at every level ($\varepsilon_{\mathrm{inv}} = \varepsilon_{\mathrm{eq}} = 0$ by construction; implementation-faithfulness verified). Injections propagate with *per-level amplification* (successive ratios 1.20, 1.32, 1.21, 1.12, 1.06, consistent with the directional gains) that *saturates* as deviations approach $\mathcal{O}(1)$ under LayerNorm renormalization. Conclusion drawn, precisely: contraction is unsupported for these models, so contraction-based uniform bounding is unavailable *here*; equally, erosion is not runaway-exponential within realistic depths — it approaches full decorrelation (≈0.9 relative state deviation) and stays there. No universal impossibility is claimed.